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The CMLA score: A novel tool for early prediction of renal replacement therapy in patients with cardiogenic shock

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ABSTRACT

Background: Early identification of cardiogenic shock (CS) patients at risk for renal replacement therapy (RRT) is crucial for improving clinical outcomes. This study aimed to develop and validate a prediction model using readily available clinical variables.

Methods: A retrospective cohort study was conducted using data from 4,133 CS patients from the MIMIC and eICU-CRD databases. Patients from MIMIC databases were randomly divided into 80 % training and 20 % validation cohorts, while those from eICU-CRD constituted the test cohort. Feature selection involved univariate logistic regression (LR), LASSO, and Boruta methods. Prediction models for RRT were developed using stepwise selection by LR and five machine learning (ML) algorithms (naive bayes, support vector machines, k-nearest neighbors, random forest, extreme gradient boosting) in the training cohort. Model performance was evaluated in both validation and test cohorts. A nomogram was constructed based on LR model. Kaplan-Meier survival analysis assessed 28-day mortality.

Results: The incidence of RRT was approximately 13 % across all cohorts. Ten variables were selected: age, anion gap, chloride, bun, creatinine, potassium, ast, lactate, estimated glomerular filtration rate (eGFR), and mechanical ventilation. Compared with ML models, the LR model showed superior predictive performance with an AUC of 0.731 in the validation cohort and 0.714 in the test cohort. Four variables that best predicted the need for RRT (age, lactate, mechanical ventilation, and creatinine) were used to generate the CMLA nomogram risk score. The CMLA model showed better predictive accuracy for RRT in the test cohort compared to the previous CALL-K model (AUC: 0.731 vs. 0.699, DeLong test $P < 0.05$). Calibration curves and decision curve analysis (DCA) indicated that the CMLA model also had good calibration (Hosmer–Lemeshow $P=0.323$) and clinical utility in the test cohort. Kaplan-Meier analysis indicated significantly higher 28-day mortality in the high-risk CMLA group.

Conclusions: A clinically applicable nomogram with four key variables was developed to predict RRT risk in CS patients. It demonstrated good performance, promising enhanced clinical decision-making.

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Introduction

Cardiogenic shock (CS) is a state of low cardiac output that results in life-threatening end-organ perfusion deficits and eventually requires urgent pharmacological and/or mechanical intervention.^{1,2} Supportive therapies for end-organ failure, including mechanical ventilation (MV) and renal replacement therapy (RRT), are often required.³

The cardiovascular and renal systems are closely linked to the pathophysiology.^{4–6} CS can lead to acute kidney injury (AKI) through various mechanisms, including inadequate perfusion, venous congestion, systemic inflammation, and release of vasoactive mediators, collectively referred to as cardiorenal syndrome type 1.⁷ Reports have shown that 13–31 % of patients with CS develop AKI and 13 % require RRT.^{1,8,9} In addition, patients with CS who require RRT have significantly higher in-hospital and long-term mortality than those who do not require RRT.^{8,10} However, although RRT is important and necessary in the treatment of patients with CS combined with severe AKI, few prediction models from previous studies have assessed the risk of RRT. The only prediction model (CALL-K score) for the risk of RRT in patients with CS lacks external validation.¹¹

In this study, we aimed to develop and validate a risk prediction model comprising commonly measured variables to predict the risk of RRT in patients with CS.

Methods

Data source and ethics statement

All relevant data for this study were extracted from the Medical Information Mart for Intensive Care III (MIMIC-III) Database CareVue subset, MIMIC-IV and the eICU Collaborative Research Database (eICU-CRD).^{12,13} MIMIC-III and MIMIC-IV contain information on patients treated at Beth Israel Deaconess Medical Center from 2001 to 2019. The eICU-CRD is a multicenter database containing data from >200,000 patients admitted to 335 ICUs at 208 hospitals across the United States between 2014 and 2015. These databases are free and publicly available data can be accessed at PhysioNet.org.¹⁴ To protect patient privacy, all personal information was de-identified and replaced with random codes. Consequently, this study did not require patient consent or ethics approval. The author completed the Collaborative Institutional Training Initiative Program (CITI) course and received approval to extract data from these databases. Analyses adhere to the transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD+AI) guidelines.¹⁵

Study population and design

Patients diagnosed with CS who underwent RRT were selected from the ICU databases. The exclusion criteria were: (1) multiple ICU admissions other than the first ICU admission, (2) age <18 years, (3) history of end-stage renal disease or receiving RRT before ICU admission, and (4) missing creatinine data. The selection process used in this study is illustrated in Supplemental Fig. 1.

A total of 2,390 patients from the MIMIC databases and 1,743 patients from the eICU-CRD database were enrolled in this study. Data from MIMIC databases were combined into the derivation cohort and randomly split into a training cohort (80 %) and a validation cohort (20 %) for the development and internal validation of the predictive model. Patients in the eICU-CRD were assigned to the external test cohort.

Data extraction

The diagnosis of CS was based on the ninth or tenth revision of the International Classification of Diseases (ICD-9/10) codes: 785.51, 998.01 for the MIMIC-III database; 785.51, R57.0, T81.11, T81.11XA, T81.11XD, T81.11XS for the MIMIC-IV database; and 785.51, R57.0 for the eICU-CRD database. PostgreSQL (version 11) and Navicat Premium (version 12) were used to extract data. Clinical data, including demographic information, vital signs, comorbidities, laboratory results, and treatments, were collected from the databases. Laboratory results were extracted within the first 24 h after ICU admission. If multiple results for clinical parameters were available during a single hospitalization, only the first result was considered. Data on MV were collected prior to RRT initiation. The CALL-K score, which includes age, hemoglobin, estimated glomerular filtration rate (eGFR), loop diuretic use, and lactate levels, was calculated based on Rodenas's study.¹¹ The primary outcome included the initiation of RRT. The secondary outcome encompassed the 28-day mortality rate.

Statistical analysis

Continuous variables were expressed as the mean (standard deviation [SD]) or the median (interquartile range [IQR]), based on the distribution pattern. Comparisons between two continuous groups were conducted using the t-test for normally distributed variables or the Wilcoxon rank sum test for non-normally distributed variables. Comparisons among three continuous groups involved the use of the ANOVA test for normally distributed variables or the Kruskal-Wallis rank sum test for non-normally distributed variables. Categorical variables were compared via the chi-squared test.

To address potential overfitting and enhance model performance, we employed univariate logistic regression (LR) ($P < 0.1$), the least absolute shrinkage and selection operator (LASSO), and the Boruta methods to select variables. The synthetic minority over-sampling technique (SMOTE) was used to address data imbalance and prevent overfitting.¹⁶ To select a model with the highest

predictive value and clinical applicability, we employed backward LR and five different ML models. The ML models included naive bayes (NB), support vector machines (SVM), k-nearest neighbors (KNN), random forest (RF), and extreme gradient boosting (XGBoost). For each algorithm, hyperparameters were optimized using the grid search method with five-fold cross-validation on the validation set to maximize the area under the receiver operating characteristic (ROC) curve (AUC). Seven evaluation metrics were utilized: AUC, sensitivity, specificity, positive predictive value, negative predictive value, F1 score, and balanced accuracy. The primary metric was AUC, and comparisons among models were made using the DeLong test. The test cohort was used to evaluate the final model's performance.

A nomogram prediction model was established based on clinical significance and regression coefficients. Consistency between predictions and actual outcomes was assessed using a calibration curve, and model calibration was evaluated with the Hosmer–Lemeshow test. Clinical utility was measured using decision curve analysis (DCA). The nomogram groups were stratified into high and low risk based on the median value. Kaplan-Meier curves were utilized to compare the 28-day mortality rates between different

Table1

Baseline characteristics of the study population.

Characteristic	Training cohort N=1913	Validation cohort N=477	Test cohort N=1743	P value
In-hospital mortality (%)	717 (37.5 %)	192 (40.3 %)	590 (34.2 %)	0.022
Gender (%)				0.041
Male	1126 (58.9 %)	291 (61.0 %)	1097 (62.9 %)	
Female	787 (41.1 %)	186 (39.0 %)	646 (37.1 %)	
Age (years)	70.7 (13.9)	70.8 (14.3)	66.0 (14.5)	<0.001
Vital signs				
SBP (mmHg)	110 (23.1)	111 (23.3)	107 (25.6)	<0.001
DBP (mmHg)	64.1 (18.3)	64.4 (18.1)	62.8 (18.7)	0.100
Heart rate (beats/min)	92.3 (21.9)	92.3 (21.6)	92.2 (22.1)	0.988
SpO ₂ (%)	95.7 (6.40)	95.4 (7.73)	95.3 (8.49)	0.246
Comorbidities (%)				
Hypertension	566 (29.6 %)	153 (32.1 %)	879 (50.4 %)	<0.001
Diabetes	695 (36.3 %)	155 (32.5 %)	542 (31.1 %)	0.003
CKD	567 (29.6 %)	118 (24.7 %)	266 (15.3 %)	<0.001
Previous MI	243 (12.7 %)	63 (13.2 %)	322 (18.5 %)	<0.001
AMI	634 (33.1 %)	142 (29.8 %)	585 (33.6 %)	0.285
COPD	534 (27.9 %)	127 (26.6 %)	266 (15.3 %)	<0.001
Malignant cancer	125 (6.53 %)	43 (9.01 %)	201 (11.5 %)	<0.001
Laboratory Outcomes				
WBC ($\times 10^9/L$)	13.8 (9.65)	13.9 (7.12)	13.4 (7.88)	0.324
Hematocrit (%)	35.0 (7.44)	35.8 (7.70)	36.3 (7.58)	<0.001
Hemoglobin (g/dL)	11.3 (2.51)	11.6 (2.61)	11.9 (2.57)	<0.001
Platelet($\times 10^9/L$)	210 [155;278]	206 [151;272]	200 [151;264]	0.015
Anion gap (mmol/L)	18.0 (5.46)	17.9 (5.34)	13.0 (5.69)	<0.001
Bicarbonate (mEq/L)	20.9 (5.31)	21.1 (5.43)	22.7 (5.48)	<0.001
Chloride (mmol/L)	102 (7.19)	102 (7.19)	102 (7.19)	0.233
Sodium (mmol/L)	137 (5.70)	138 (5.18)	137 (5.85)	0.123
Potassium (mmol/L)	4.51 (0.98)	4.51 (0.95)	4.34 (0.90)	<0.001
Calcium (mg/dL)	7.42 (2.70)	7.57 (2.88)	8.50 (0.97)	<0.001
Albumin(g/dL)	3.19 (0.61)	3.24 (0.62)	3.07 (0.65)	<0.001
Glucose(mg/dL)	155 [118;221]	151 [114;214]	147 [114;215]	0.024
Bun (mg/dL)	29.0 [19.0;47.0]	30.0 [19.0;48.0]	26.0 [17.0;42.0]	<0.001
Creatinine (mg/dL)	1.40 [1.00;2.10]	1.40 [1.00;2.00]	1.35 [0.99;1.99]	0.293
Ckmb(U/L)	12.0 [4.00;49.5]	12.0 [4.00;43.8]	9.55 [3.29;45.8]	0.033
ALT (U/L)	47.0 [22.0;134]	50.0 [22.0;167]	45.0 [24.0;138]	0.653
AST (U/L)	88.0 [37.0;271]	85.0 [35.0;303]	68.0 [30.0;225]	<0.001
Bilirubin (mg/dL)	0.80 [0.50;1.40]	0.80 [0.50;1.40]	0.80 [0.50;1.33]	0.305
Lactate (mmol/L)	2.60 [1.60;4.60]	2.50 [1.60;4.50]	3.51 [1.80;7.00]	<0.001
PT (s)	15.2 [13.1;19.6]	15.4 [13.3;21.6]	15.3 [13.1;19.7]	0.169
APTT (s)	36.3 [28.8;57.7]	36.4 [30.0;55.3]	34.4 [29.0;46.0]	<0.001
Inr	1.40 [1.20;1.90]	1.40 [1.20;2.10]	1.30 [1.10;1.70]	<0.001
eGFR (mL/min/1.73m ²)	46.7 [29.5;71.5]	47.2 [28.5;72.6]	51.7 [31.0;75.8]	<0.001
Treatments (%)				
IABP	360 (18.8 %)	73 (15.3 %)	395 (22.7 %)	<0.001
MV	1161 (60.7 %)	281 (58.9 %)	1046 (60.0 %)	0.760
RRT	250 (13.1 %)	62 (13.0 %)	215 (12.3 %)	0.790
Loop diuretic (%)	342 (17.9 %)	79 (16.6 %)	78 (4.48 %)	<0.001
CALL-K score	3.27 (1.61)	3.35 (1.68)	3.45 (1.61)	0.027

Abbreviations: RRT, Renal Replacement Therapy; SBP, systolic blood pressure; DBP, diastolic blood pressure; SpO₂, pulse oxygen saturation; WBC, white blood cell; CKD, chronic kidney disease; AMI, acute myocardial infarction; COPD, chronic obstructive pulmonary disease; AST, aspartate aminotransferase; ALT, alanine transaminase; PT, prothrombin time; APTT, activated partial thromboplastin time; IABP, intra-aortic balloon pump; MV, mechanical ventilation

groups, with comparisons conducted using the log-rank test.

Because this was a hypothesis-generating epidemiological study, we did not estimate the sample size. Variables with missing values of $\geq 25\%$ were excluded, while the KNN method was used to impute missing values $< 25\%$. Data were analyzed using R (version 4.4.1; R Foundation for Statistical Computing), and a two-sided P-value < 0.05 indicated statistical significance.

Results

Baseline patient characteristics

After screening, 1913 patients in the training cohort, 477 patients in the validation cohort, and 1743 patients in the test cohort were enrolled in this study. The occurrence of RRT use was 13.1 %, 13.0 %, and 12.3 % in the training, validation, and test cohorts, respectively ($P = 0.790$). The in-hospital mortality rate was 37.5 % in the training cohort, 40.3 % in the validation cohort, and 34.2 % in the test cohort ($P = 0.022$). Baseline patient characteristics are detailed in [Table 1](#). Comparisons within RRT groups for the derivation and test cohorts are included in Supplementary Table 1 and Supplementary Table 2.

Feature selection

Based on LR, LASSO, and Boruta methods, we ultimately identified 10 variables: age, anion gap, chloride, bun, creatinine, potassium, ast, lactate, eGFR, and MV ([Fig. 1](#)). The specific variable selection results can be found in Supplementary Fig. 2 and Supplementary Table 3.

Performance of prediction models

The prediction performance metrics of each model is shown in [Table 2](#). Upon completion of cross-validation, AUCs were computed for LR and each ML model in the validation cohort: the LR model yielded an AUC of 0.731 (DeLong test, reference); NB, 0.687 (DeLong

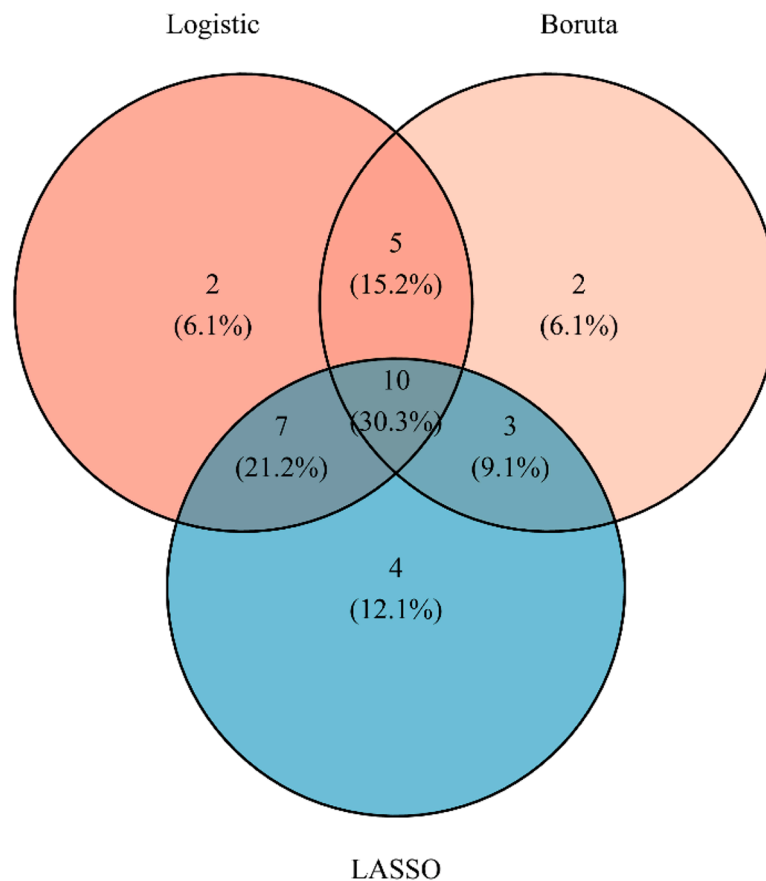


Fig.1. Screening of characteristic predictors.

Abbreviations: LASSO, least absolute shrinkage and selection operator.

Table2

The prediction performance of each model.

Algorithm used in model development	LR	NB	SVM	KNN	RF	XGBoost
Derivation cohort						
Training cohort						
AUC	0.764	0.750	0.767	0.817	0.892	0.874
Validation cohort						
AUC	0.731	0.687	0.702	0.683	0.684	0.718
SE	0.468	0.387	0.532	0.484	0.419	0.403
SP	0.839	0.853	0.795	0.752	0.795	0.812
PPV	0.302	0.282	0.280	0.226	0.234	0.243
NPV	0.913	0.903	0.919	0.907	0.902	0.901
Recall	0.468	0.387	0.532	0.484	0.419	0.403
F1 score	0.367	0.327	0.367	0.308	0.301	0.303
Balanced Accuracy	0.653	0.620	0.664	0.618	0.607	0.608
DeLong test	Reference	0.036	0.071	0.036	0.023	0.556
Testing cohort						
AUC	0.714	0.678	0.691	0.680	0.689	0.650
SE	0.498	0.340	0.423	0.507	0.442	0.540
SP	0.801	0.868	0.822	0.748	0.783	0.693
PPV	0.260	0.265	0.251	0.221	0.222	0.198
NPV	0.919	0.903	0.910	0.915	0.909	0.915
Recall	0.498	0.340	0.423	0.507	0.442	0.400
F1 score	0.342	0.298	0.315	0.307	0.296	0.290
Balanced Accuracy	0.649	0.604	0.623	0.628	0.612	0.616
DeLong test	Reference	0.005	0.003	0.010	0.013	<0.001

Abbreviations: LR, logistic regression; NB, naive bayes; SVM, support vector machine; KNN, k-nearest neighbors; RF, random forest; XGBoost, extreme gradient boosting; AUC, area under the curve; SE, sensitivity; SP, specificity; PPV, positive predictive value; NPV, negative predictive value

test, $P = 0.036$); SVM, 0.702 (DeLong test, $P = 0.071$); KNN, 0.683 (DeLong test, $P = 0.036$); RF, 0.684 (DeLong test, $P = 0.023$); and XGBoost, 0.718 (DeLong test, $P = 0.556$) (Fig. 2A). In the test cohort, the AUCs were as follows: LR, 0.714 (DeLong test, reference); NB, 0.678 (DeLong test, $P = 0.005$); SVM, 0.691 (DeLong test, $P = 0.003$); KNN, 0.680 (DeLong test, $P = 0.010$); RF, 0.689 (DeLong test, $P = 0.013$); and XGBoost, 0.650 (DeLong test, $P < 0.001$) (Fig. 2B). The DeLong test showed that the AUC of the LR model was significantly higher than that of other ML models in the test cohort ($P < 0.05$).

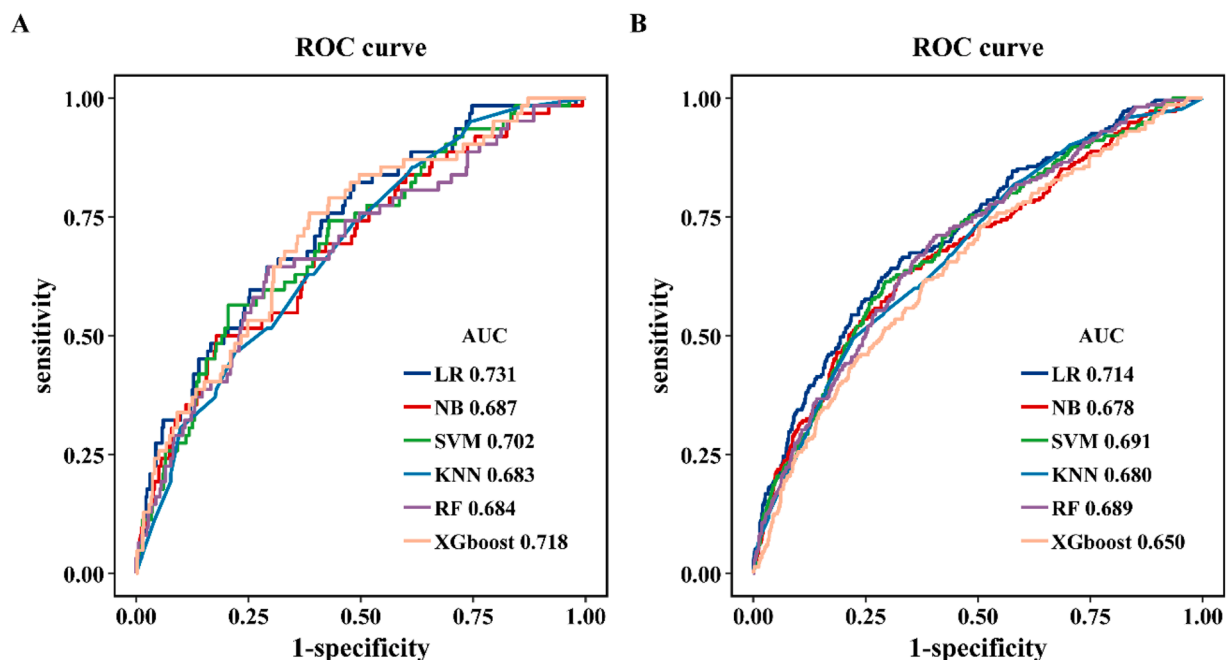


Fig. 2. Receiver operating characteristic (ROC) curves for the prediction of renal replacement treatment. (A) ROC curves in the validation cohort. (B) ROC curves in the test cohort.

Abbreviations: LR, logistic regression; NB, naive bayes; SVM, support vector machine; KNN, k-nearest neighbors; RF, random forest; XGBoost, extreme gradient boosting; AUC, area under the curve.

Assessment of nomogram

To eliminate collinearity among variables and to ensure clinical significance and interpretability, four variables were included to construct a LR model: age, lactate, MV, and creatinine. We also developed a nomogram based on the LR model, named CMLA (Fig. 3). In comparison with the previously established CALL-K score, the AUCs in the derivation cohort were 0.768 (95 % CI, 0.742–0.793) for CMLA and 0.686 (95 % CI, 0.654–0.718) for CALL-K (Fig. 4A). In the test cohort, the AUC values were 0.731 (95 % CI, 0.696–0.767) for CMLA and 0.699 (95 % CI, 0.661–0.736) for CALL-K (Fig. 4B). The DeLong test demonstrated that the AUC of the CMLA was significantly higher than that of CALL-K ($P < 0.05$). Furthermore, the calibration curve demonstrated good consistency between the predicted and actual observations in both the derivation cohort (Hosmer–Lemeshow test p -value = 0.058) and the test cohort (Hosmer–Lemeshow test p -value = 0.323) (Fig. 4C, D). Additionally, the DCA curves demonstrated that the CMLA had good clinical benefit in both the derivation cohort and the test cohort (Fig. 4E, F).

Subgroup analysis of the CMLA also demonstrated good performance, with AUCs ranging from 0.683 to 0.781 (Supplemental Table 4). Only in the acute myocardial infarction (AMI) subgroup of the test cohort was the AUC poor at 0.683, whereas in other subgroups, it was above 0.700. The KM curves demonstrated that the mortality rate in the RRT group was significantly higher than that in the non-RRT group (Fig. 5A, B). Stratification of the CMLA score by the median further demonstrated that the 28-day mortality in the high-score group was significantly higher than that in the low-score group (Fig. 5C, D), with a log-rank test result of $P < 0.001$.

Discussion

Early identification of patients with CS who require RRT is crucial for clinical management. In this multicenter retrospective study, LR and ML models were applied to construct several models aimed at predicting RRT use in patients with CS. These models underwent external validation, ultimately identifying LR as having superior predictive performance. CMLA model based on LR consists of four readily available indicators: age, lactate, MV, and creatinine, which may serve as a valuable tool for identifying and managing patients with CS who are at high risk of requiring RRT. In addition, an association was found between 28-day mortality and increasing CMLA score.

To the best of our knowledge, the CALL-K score developed by Rodenas et al. is the only existing prediction model for RRT use in patients with CS.¹¹ This system employs hemoglobin, age, lactate, pre-admission loop diuretic use, and eGFR to estimate the risk of RRT in CS with good predictive performance. However, it was developed in a single-center study without external validation. In our study, we compared the CMLA score with the CALL-K score, and found that the CMLA score showed superior predictive accuracy for the need for RRT in both the derivation and test cohorts. Notably, hemoglobin and pre-admission loop diuretic use did not exhibit significant independent predictive value in our study. Variations in study population characteristics and the severity of their medical conditions may have contributed to these discrepancies. The patient population in our study include a higher proportion of elderly, as well as a higher prevalence of chronic kidney disease and AMI. Additionally, our cohort had a higher frequency of MV but relatively

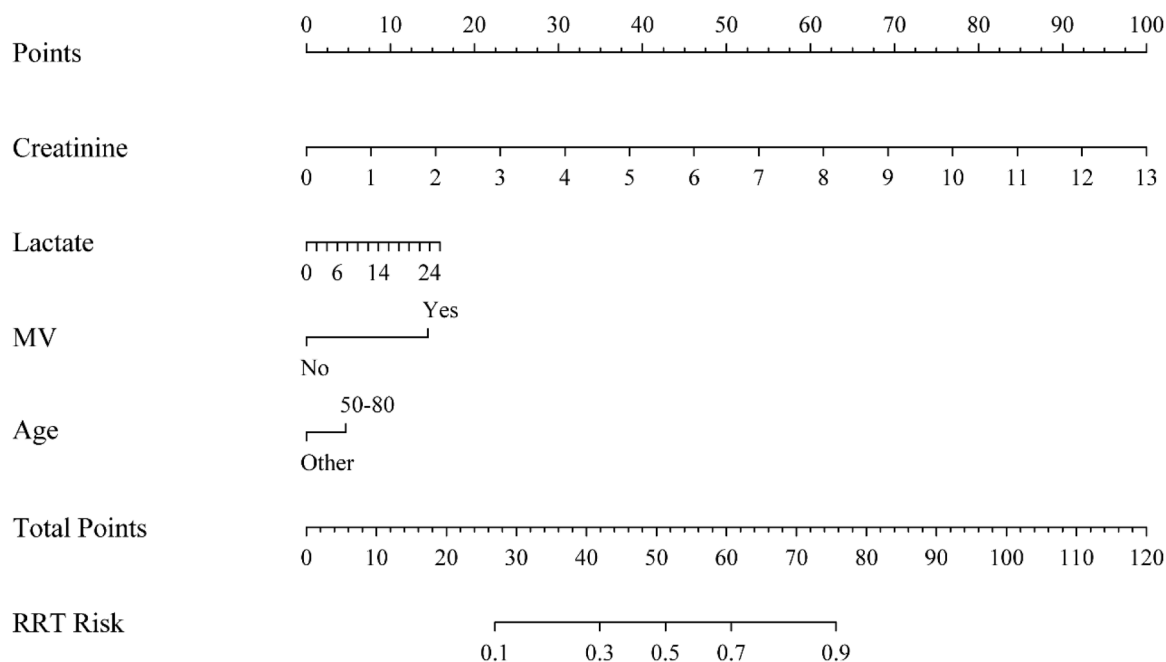


Fig. 3. Nomogram for the prediction of renal replacement treatment in cardiogenic shock patients. Abbreviations: MV, mechanical ventilation; RRT, renal replacement treatment.

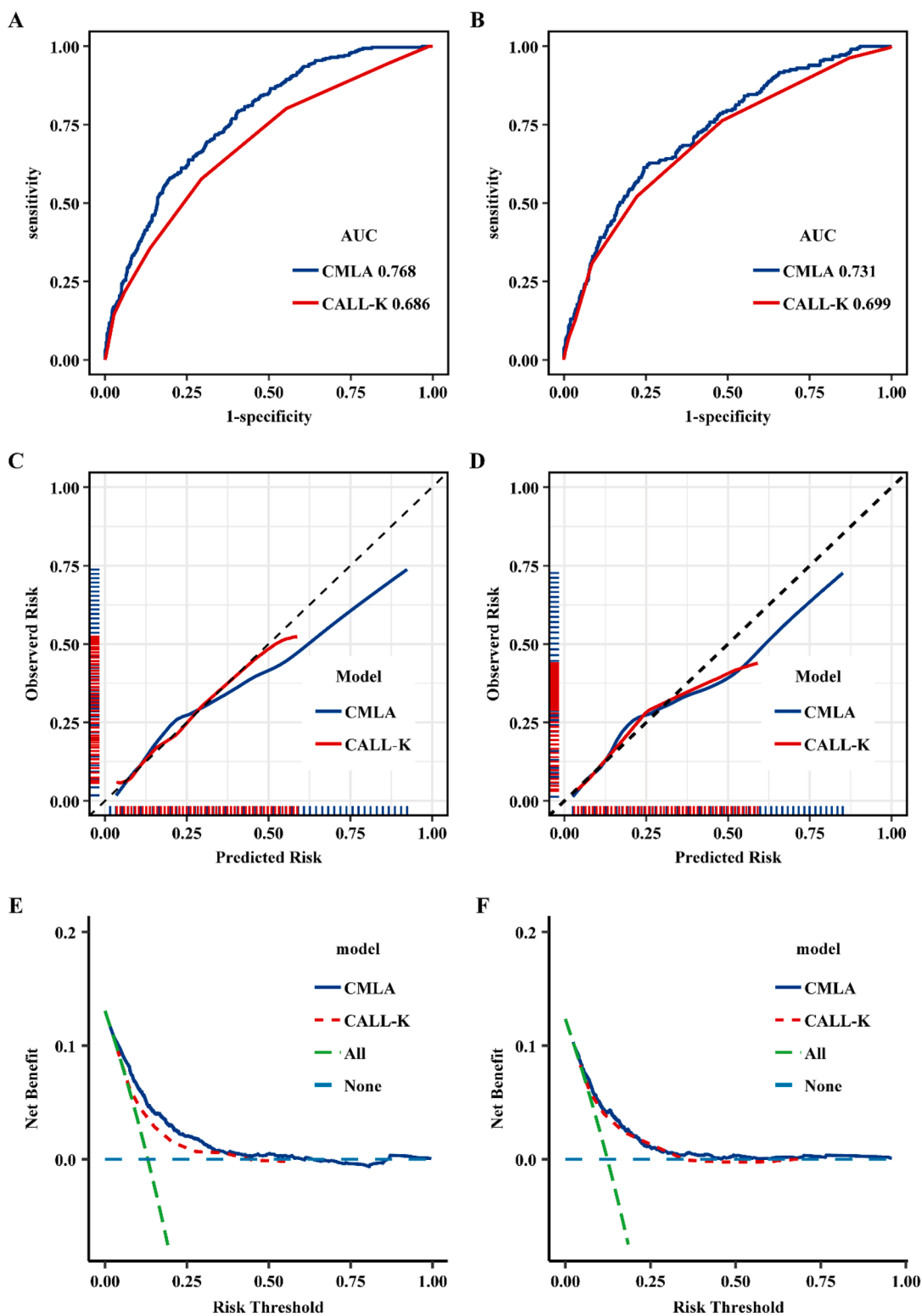


Fig. 4. Model prediction performance and clinical usefulness. (A) ROC curves in the derivation cohort. (B) ROC curves in the test cohort. (C) Calibration curves in the derivation cohort. (D) Calibration curves in the test cohort. (E) Decision curves analysis in the derivation cohort. (F) Decision curves analysis in the test cohort.

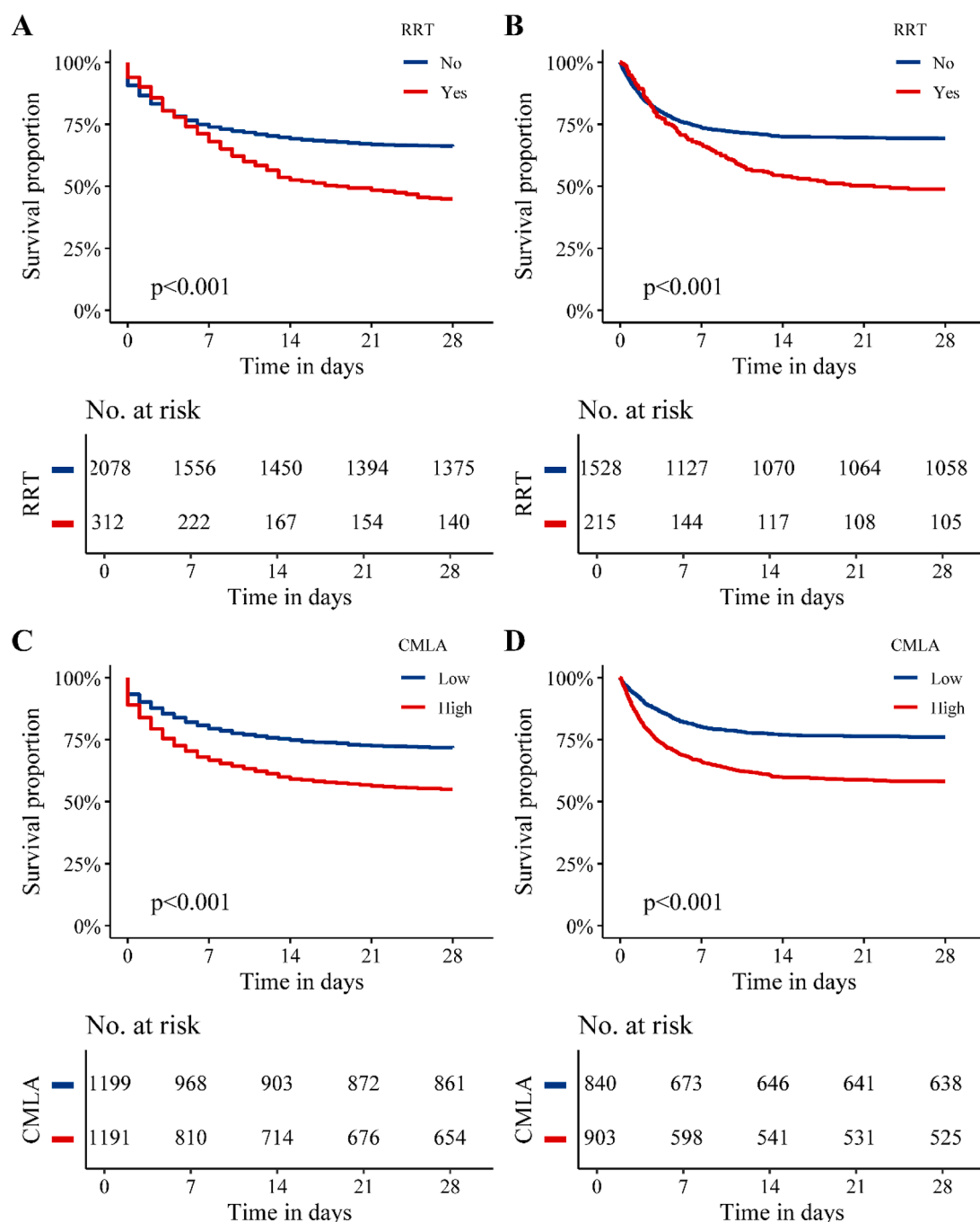


Fig. 5. The Kaplan-Meier (KM) curves for 28-day survival proportion. (A) The KM curve of the derivation cohort based on renal replacement treatment (RRT). (B) The KM curve of the test cohort based on RRT. (C) The KM curve of the derivation cohort based on the median of the CMLA score. (D) The KM curve of the test cohort based on the median of the CMLA score.

less use of intra-aortic balloon pump (IABP). These factors may have contributed to divergent analytical results.

The creatinine level is the most commonly used clinical indicator of renal function.^{17,18} However, in the early stages of CS, creatinine levels may not show a significant increase, limiting their predictive sensitivity for RRT requirement. Therefore, a combined predictive model may enhance the accuracy and reliability of prognostication. The variables in our predictive model are readily available and have been associated with AKI in previous studies. Elevated lactate levels are common in CS patients, indicating organ hypoperfusion and poor prognosis.¹⁹ Age is closely related to prognosis, with individuals aged 50-80 at higher risk due to the U-shape relationship between age and RRT utilization.¹¹ RRT may not be employed in very elderly adults who have a poor prognosis in CS. In

addition, our study identified a close association between MV and the occurrence of RRT. The respiratory and renal systems are the two most commonly affected organs in CS patients with multi-organ failure.^{20,21} On the one hand, patients with CS and consequent pulmonary edema or severe respiratory impairment usually require MV.²² On the other hand, MV affects systemic and renal hemodynamics through its impact on cardiac output.^{23,24} Additionally, MV may induce a biotrauma-pulmonary inflammatory response, leading to the release of systemic inflammatory mediators.²⁵ Previous studies have established a close relationship between MV and AKI.²⁶ A meta-analysis of 23 studies involving 10,333 patients demonstrated a three-fold increase in AKI in critically ill patients receiving invasive MV.²⁷ Our findings provide evidence that MV continues to be a robust risk factor for RRT in patients with CS.

Our predictive model is displayed as a nomogram, which facilitates implementation and simplifies the assessment for clinicians to determine whether a CS patient requires RRT. Our model not only demonstrates good accuracy and calibration but also exhibits clinical applicability in both internal and external validations. Moreover, the KM analysis indicates a significant elevation in 28-day mortality among patients with a nomogram score surpassing the median value. This observation implies that patients identified as high-risk by the nomogram could potentially derive greater clinical advantages from receiving early RRT treatment. Furthermore, it is noteworthy that although the timing of RRT remains controversial, its initiation is imperative whenever life-threatening changes in fluid, electrolyte, and acid-base balance occur.^{28–30}

Although our predictive model has demonstrated efficacy in early prediction of RRT for patients with CS, optimizing the indicators remains challenging. Previous studies have identified indicators associated with AKI, including elevated central venous pressure (CVP), decreased mean arterial pressure (MAP), and nephrotoxic drug usage. In patients with CS, elevated CVP is often observed due to compromised cardiac function, which numerous studies have shown to decrease glomerular filtration rate.^{31,32} Furthermore, the maintenance of a specific MAP is a therapeutic goal for patients with CS, as peripheral organ perfusion is frequently compromised in this population. A multicenter study demonstrated that low MAP at the initiation of RRT correlated with increased mortality, particularly when MAP levels were below 82.7 mmHg.³³ Another study showed that a MAP of less than or equal to 59 mmHg was independently associated with the progression of AKI.³⁴ If high doses of vasoactive drugs are insufficient to maintain blood pressure levels, mechanical assistance should be promptly initiated.³⁵ Additionally, when patients have sepsis and septic shock, antibiotic dosage should be closely adjusted according to renal function to reduce nephrotoxicity.

Our study has several strengths. First, it was a multi-center, large-sample study utilizing data from extensive databases. Second, our research includes analyses that demonstrate both internal and external validation. Finally, all predictors were available within a relatively short period after admission, allowing clinicians to evaluate the risk of RRT early in the hospitalization process.

The study has several limitations. First, this study was based on two public critical care databases in the United States, which may introduce some selection bias. Second, as this is a retrospective study, these findings should be interpreted with caution, and further validation is needed in a prospective multi-center study.

In conclusion, we developed and validated a simple and reliable model for predicting the risk of RRT in patients with CS. The model demonstrated good predictive accuracy, discrimination, and clinical utility, indicating good performance for clinical applications, thereby aiding clinicians in predicting the risk of RRT in patients with CS and enhancing clinical management.

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Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

CRedit authorship contribution statement

Shuo Pang: Formal analysis, Data curation, Writing – original draft. **Shen Wang:** Formal analysis, Data curation, Writing – review & editing. **Chu Fan:** Formal analysis. **Fadong Li:** Data curation, Writing – review & editing. **Wenxin Zhao:** Formal analysis. **Boqun Shi:** Data curation, Writing – review & editing. **Yue Wang:** Formal analysis. **Xiaofan Wu:** Methodology, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cpcardiol.2024.102870](https://doi.org/10.1016/j.cpcardiol.2024.102870).

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